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Wealth, homeownership, and land use regulatory changes

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Abstract

This paper examines patterns of land use regulatory changes in Santiago, Chile, focusing on downzoning (stricter regulations) and upzoning (more flexible regulations), and their associations with sociodemographic factors. We analyze an original georeferenced dataset covering all 34 urban municipalities of Santiago over 20 years, together with 19 semi-structured interviews with key actors. The quantitative analysis suggests that high-income neighborhoods are associated with a greater likelihood of downzoning when homeownership is comparatively low. With our block-level measure of wealth, however, this association is not statistically distinguishable from zero at the sample-average homeownership rate and becomes negative at higher rates. Therefore, in affluent neighborhoods, the significant presence of homeowners is associated with less probability of downzoning. Outside high-income neighborhoods, higher homeownership is instead associated with downzoning. Qualitative evidence points to the growing role of organized residents, with socially connected groups using participatory reforms and informal ties to municipal decision-makers to shape local regulatory debates. This positive association is absent when the analysis is restricted to large-scale zoning changes. The findings contribute to debates between the "growth machine" and "homevoter" theories by suggesting that the relevance of each framework may vary across patterns of social segmentation and local demographic structures.

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Land use regulation, homeownership, upzoning, downzoning, affordable housing

摘要

本文考察智利圣地亚哥的土地利用管制变动格局，重点分析降容区划（管制更为严苛）与提容区划（管制更为灵活）及其与社会人口因素之间的关联。我们分析一套覆盖圣地亚哥全部 34 个市政辖区、时间跨度 20 年的自建地理配准数据集，以及对核心行动者开展的 19 场半结构化访谈。定量分析表明，在住房自有率相对较低的情况下，高收入街区更有可能经历降容区划。然而，采用本文地块级财富测度后，当住房自有率取样本均值时，该关联在统计上与 0 无显著差异；当住房自有率进一步升高时，该关联转为负向。因此，在富裕街区中，住房自有者占比较高与降容区划的发生概率更低存在关联。在非高收入街区，较高的住房自有率反而与降容区划存在关联。质性证据显示，有组织居民的作用日渐凸显，社会联结紧密的群体依托参与式改革以及与市政决策者的非正式联结，影响本地管制议题的讨论。当分析限定于大规模的区划调整时，这种正向关联便不复存在。研究结果表明，各理论框架的解释力会随社会分异格局和本地人口结构的差异而变化，为“增长机器”理论与“房产选民”理论之间的争论提供了佐证。

关键词

土地利用管制、住房自有率、提容区划、降容区划、经济适用住房

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Introduction

Why do some urban neighborhoods become more restrictively regulated over time while others are open to denser development? A foundational literature shows that stringent land use controls constrain housing supply and raise prices (Brueckner, 2009), especially in high-demand cities, and that such restrictions reflect not only planning goals but also the distributional interests of incumbent residents (Brouwer and Trounstine, 2024; Glaeser and Ward, 2009; Hilber and Robert-Nicoud, 2013; Hilber and Vermeulen, 2016; Kok et al., 2014). More recent work argues that these dynamics contribute directly to today's housing affordability crisis, disproportionately affecting middle- and low-income groups (Anagol et al., 2021; Been et al., 2019; Brueckner, 2009; Gyourko and Molloy, 2015; Hills and Schleicher, 2015; Manville et al., 2020), though upzoning alone may not reduce urban inequality where deeper structural issues persist (Rodríguez-Pose and Storper, 2020; Serkin, 2020). Yet less is known about the mechanisms that lead some local governments to favor stricter regulations while others permit densification, and whether and how the socioeconomic composition of neighborhoods shapes those outcomes (Brouwer and Trounstine, 2024; Cheung et al., 2024).

This paper examines whether neighborhood sociodemographic characteristics and urban attributes are associated with changes in urban land use regulations. We leverage a novel dataset, which, to our knowledge, is the most comprehensive effort to codify and

geo-reference local land use regulations at the lot level in a major metropolitan area over two decades. We focus on the 34 urban municipalities that constitute the Metropolitan Area of Santiago, Chile, home to around 7 million people—approximately 40% of the national population. Our dataset records all zoning and ordinance changes between December 2002 and October 2019 and includes evidence from 19 semi-structured interviews with policymakers, consultants, real estate professionals, and NGOs.

We employ a policy-oriented mixed-methods approach (Richwine et al., 2022). Our quantitative analysis examines two possible lot-level outcomes over the study period: stricter regulations preventing dense real estate development ("downzoning"), or more flexible regulations allowing denser construction ("upzoning"). Unlike most studies focusing on the maximum floor area ratio (FAR) (see, e.g., Anagol et al., 2021; Been et al., 2014; Büchler and Lutz, 2024), we analyze both FAR and building height limits, the norms our interviews identified as most important. We enrich our dataset by incorporating administrative sources, including census data, and estimate a linear probability model of upzoning or downzoning. Across various specifications, we examine how regulatory changes are associated with baseline socioeconomic and infrastructure variables, focusing on homeownership, high-income residents, and transport accessibility.

Our key finding is that high-income neighborhoods are associated with a higher likelihood of downzoning in areas with low homeowner presence throughout the 20-year study period. At the sample-average homeownership rate, the block-level association is statistically indistinguishable from zero, and it reverses in affluent areas with high homeownership; under the coarser municipality-level measure, the reversal occurs at lower homeownership rates. This suggests that property ownership counterbalances the relationship between high-income residents and stricter regulations. However, in non–high-income areas, homeownership is positively associated with the gradual tightening of land use regulations over time. Our analysis does not reveal consistent associations between these sociodemographic factors and upzoning.[1]

We complement these findings with 19 interviews with key Chilean land use stakeholders. The interviews reveal two types of homeowners who push municipal councils in opposite directions: those seeking to profit from land sales and those aiming to preserve the neighborhood's character. Profit-driven advocacy for upzoning appears concentrated in high-income areas. Across Santiago, residents are increasingly organized and active in zoning debates, and recent participation reforms tend to benefit socially connected residents who use formal and informal channels to push for stricter regulations.

We also examine two additional questions. First, following U.S. literature suggesting that preferences for land use status quo are particularly strong in single-family areas (Ellickson, 2020), we test whether this pattern holds in Santiago. It does: the association between high-income residents and downzoning is also present in predominantly single-family zones; it weakens as homeownership rates increase; and in non-wealthy single-family zones, homeownership remains associated with downzoning. Second, we explore whether large-scale regulatory changes alter this dynamic, as prior research suggests the real estate industry tends to oppose broad downzoning measures (Hills and Schleicher, 2015). Here we find no clear positive association between high-income residents and downzoning; through 2015 it is negative at average and higher homeownership rates, while the link between homeownership and downzoning remains strong outside high-income areas.

Our paper contributes to the debate between two urban governance theories: the "growth machine" model, where real estate interests drive city policymaking (Molotch, 1976), and the "homevoter" hypothesis, which emphasizes homeowners' influence (Been et al., 2014; Fischel, 2001; Gabbe, 2018). We argue that both theories are plausible depending on a city's social segmentation.

Our analysis also advances the literature by examining these dynamics outside the Anglo-American planning systems that dominate existing scholarship. Unlike discretionary systems such as the United States, where project-by-project decisions give organized residents ample opportunity to block individual developments (Schleicher, 2013), Chile operates under a formalized, plan-based system requiring structured regulatory procedures (Hirt, 2012). This contrast lets us assess whether the socioeconomic dynamics identified in that literature operate similarly when the rules of the game differ substantially.

Methodologically, we contribute the first comprehensive dataset of land use regulatory changes in Santiago, obtained directly from local ordinances and zoning plans. Prior studies often rely on surveys (Gyourko et al., 2021; Hilber and Vermeulen, 2016; Jackson, 2016), focus solely on FAR, lack temporal depth, or analyze a single municipality (Been et al., 2014; Brueckner et al., 2017; Brueckner and Singh, 2020; Gabbe, 2018; Schuetz, 2009; Zhang, 2023).

Theories and evidence on land use regulatory changes

The dominant early theory, proposed by Molotch (1976), posits that local elites and the real estate sector form coalitions to promote urban growth, influencing policymakers through lobbying and campaign contributions (Logan and Molotch, 1987). By contrast, Fischel (2001)'s "homevoter hypothesis" argues that homeowners drive land use policies to protect property values, as they dominate local elections. This theory was primarily applied to U.S. suburbs, where homogeneous communities face low collective action costs.

Recent empirical evidence suggests that homevoter dynamics shape land use politics even in large cities. Studies in New York (Been et al., 2014) and Los Angeles (Gabbe, 2018) find that higher homeownership rates reduce upzoning probabilities. Research on civic participation shows that homeowners engage more in local planning commissions, often opposing new housing (Einstein et al., 2019). Hankinson (2018) finds that even renters in high-priced cities resist nearby development. Schleicher (2013) attributes big-city restrictions to fragmented, individualized decision-making that lets local opposition prevail over industry interests, while Ellickson (2020) argues that homeowners prefer the status quo due to perceived downsides of increased density.

Despite extensive research on homeowners' influence, little is known about whether their impact varies by socioeconomic status. Given the deep inequalities in major cities (Florida, 2017), we address this gap by analyzing whether residents' income status is associated with zoning changes.

Our study also contributes to a growing Chilean literature on the social and political tensions surrounding land use regulation and neighborhood change. Qualitative work documents how municipalities navigate competing pressures from organized residents and developers (Gil and Bucarey, 2021; Ubilla-Bravo, 2020; Wellington, 2020), while ethnographic studies tie identity, nostalgia, and belonging to mobilization against urban change (Colin, 2017, 2021; Colin and Iturrieta, 2022; Márquez and Pérez, 2008).

The most systematic quantitative study to date finds that responses to densification in central Santiago are often delayed and fragmented, though based on only 20 zoning areas (Vicuña and Urbina, 2023). We expand this literature with a mixed-methods approach combining a comprehensive dataset of all zoning changes across Santiago's urban municipalities with stakeholder interviews, enabling us to trace regulatory change over time and analyze how its direction varies with residents' sociodemographic profiles.

Research design

This study adopts a mixed-methods approach, combining quantitative regression analysis and qualitative interviews to explore the interplay between sociodemographic factors, urban attributes, and land use regulations across all 34 urban municipalities of the Santiago Metropolitan Area.

Case study

This paper focuses on Santiago, the capital of Chile. Despite decades of economic growth and falling poverty, housing remains a major policy challenge. The quantitative housing deficit has stayed high, falling to about 480,000 units in 2015 before rising to around 550,000 units by 2022 (Ministerio de Desarrollo Social, 2023). Most affected households share overcrowded dwellings ("allegamiento"), citing unaffordable rents as the main reason (Ministerio de Desarrollo Social, 2018). Moreover, households in informal settlements have tripled over the past decade to around 120,000 (Centro de Estudios Techo-Chile, 2025). Meanwhile, housing prices have increased threefold relative to wages (Vergara-Perucich and Aguirre-Nuñez, 2019), weakening government subsidies' effectiveness (Razmilic Burgos, 2010; Gil, 2019).

In Chile, urban land use regulations are established through a structured planning process governed by national laws and executed at both regional and municipal levels. Municipalities hold the authority to develop local regulatory plans ("planes reguladores comunales"), which define zoning, building height limits, densities, and land uses within their jurisdictions. These plans are expected to balance city growth with public needs, address environmental and historical considerations, and guide urban development in alignment with legal and technical standards. However, plan-making is politically sensitive, as it must reconcile the public interest with private property rights, ensuring neither undue influence from interest groups nor excessive restrictions that could harm homeowners or reduce overall welfare.

The legal framework—principally the General Law and Ordinance of Urbanism and Constructions—sets out a detailed, multi-stage process for creating or modifying local plans, including community consultation, transparency requirements, environmental assessments, and approval by the Municipal Council. These robust procedures are also demanding: developing a local regulatory plan takes six years on average, with many municipalities depending on central-government support, producing significant disparities in how up-to-date local plans are (Ministerio de Vivienda y Urbanismo, 2013).

Chile offers an interesting contrast to countries with more discretionary land use systems. In the U.S., decisions often occur project by project, enabling local homeowners to block changes more easily than broader groups like housing consumers or developers (Schleicher, 2013). Some argue that shifting to less discretionary, comprehensive planning could boost housing supply by increasing predictability and inclusiveness (Hills and Schleicher, 2015), while others defend discretion as a means of mediating competing interests (Rose, 1983).

Table 1. Changes in regulation over the 2002–2019 period.

			Max. Height (m)			Max. FAR		
Category	Total area (sq. km.)	Obs.	2002	2019	Diff.	2002	2019	Diff.
Not rezoned	68.93	6444	12.71	12.71	0.00	1.06	1.06	0.00
Downzoned	30.30	5473	12.35	11.09	−1.26	1.89	1.38	−0.51
Upzoned	70.48	7716	9.36	12.55	3.19	0.73	1.47	0.74

Note: This table shows the total area of the sample that was upzoned, downzoned, and not rezoned between 2002 and 2019, excluding all lots that at some point experienced a mixed change. It also shows the weighted average of the building height limit and maximum allowable FAR for each type, using lot sizes as weights for 2002 and 2019. To compute the averages, we excluded the observations that did not have a numeric value for the maximum permitted height or FAR (as opposed to unlimited). Supplemental Table B.1 in the Appendix details all possible cases.

Comparable contrasts appear elsewhere: the U.K. relies on case-by-case approvals, whereas France and Germany guarantee the right to build under adopted plans (Hirt, 2012). Chile aligns with this latter model, as any modification must follow formal procedures, limiting ex-post opposition that can block densification in more discretionary systems.

Study samples and data

Regulation data collection. We constructed a novel database of municipal land use regulations, focusing on zoning plans and ordinances enacted between 2002 and 2019, for all 34 urban municipalities in the Santiago Metropolitan Area (about 6.7 million inhabitants in the 2017 National Census).[2] The project began in collaboration with TOCTOC, a Chilean real estate platform, but due to gaps in their data, we organized a team of research assistants—several of them trained geographers—who, under our supervision, systematically collected all local ordinances and zoning plans, including their subsequent amendments, for every urban municipality in Santiago. Most ordinances were obtained from the *Official Gazette of the Republic of Chile* (*Diario Oficial*) and the Library of the National Congress, while zoning plans were retrieved from multiple sources, including the Ministry of Housing and Urbanism, the National Archive, and municipal planning departments. To the best of our knowledge, this process captured all local ordinances and their modifications during the study period. We then manually extracted all land use provisions into standardized sheets for each municipality. From these rules and the zoning plans, we georeferenced the data and built shapefiles that map the regulations governing every lot in the study area. We track two outcomes, upzoning and downzoning (defined below), in a balanced panel.

Our sample excludes areas designated exclusively for commercial or industrial use, as the dynamics under study are specific to residential zones; areas with distinctive land use regulations, such as heritage-protected zones, which follow different legal frameworks; and areas that lacked a local building height limit or a maximum allowable FAR in 2002, since our focus on local regulatory decisions presupposes the existence of such rules. Still, our dataset covers most residential areas in Santiago (169 square kilometers). This figure is the sum of the not-rezoned, downzoned, and upzoned areas in Table 1; lots whose height and FAR limits moved in opposite directions are classified as mixed changes and excluded from Table 1 and the analysis, for the reason given below.

Table 2. Descriptive statistics for census blocks in the study area (2002).

Variable	Average	Standard Deviation	Min.	Max.
Share of wealthy households	0.27	0.28	0.00	1.00
Homeownership rate	0.71	0.18	0.00	1.00
Number of households	41.52	54.25	1.00	1,109.00
Number of dwellings	41.00	59.03	2.00	1,201.00
Distance to subway (km)	3.75	2.37	0.04	12.80
Surface area [m^2]	14,089	44,789	75.25	1,910,453

Note. This table summarizes block-level descriptive statistics using data from the 2002 Census.

Regulatory changes. Our analysis focuses on two key land use regulations: building height limit and the maximum allowable FAR. We selected these rules based on qualitative evidence collected during the study. Interviewees consistently highlighted FAR as the most influential regulation for the real estate industry. However, as FAR is not always well understood by the public, height limits are often the more contentious norm, since they visibly convey potential neighborhood transformations.

We coded the building height limit and maximum allowable FAR (*coeficiente de constructibilidad*), the main instruments shaping residential development in Santiago (Vicuña and Moraga, 2021), using early 2002 as the baseline period. We categorized changes between 2002 and 2019 as upzoning, downzoning, no change, or mixed. Upzoning refers to regulations becoming more permissive: an increase in the height limit, the maximum FAR, or both, with no decrease in the other. Downzoning is the opposite: a reduction in development capacity through a decrease in the height limit, in the maximum FAR, or in both. The mixed category, simultaneous increases in one parameter and decreases in the other, is excluded because its direction is difficult to interpret, a limitation we acknowledge. Descriptive statistics for these changes are provided in Table 1.[3]

Key variables. Our two key variables are block-level wealth and homeownership rates. The homeownership rate is available from Chile's 2002 Population and Housing Census, but wealth is not, so we use the Socio-Material Territorial Index (ISMT) of the Pontificia Universidad Católica de Chile (Observatorio de Ciudades UC, 2022). It combines head-of-household education, housing quality, overcrowding, and co-residence into a weighted score classifying households into three socioeconomic groups. The highest group is labeled ABC1, following the conventional Chilean socioeconomic classification, which ranks households from ABC1 (highest) down to E (lowest). ABC1 is therefore a composite socioeconomic stratum rather than an income percentile or an income decile; it comprises the top 14.1% of the population nationally.

We define a block as wealthy when more than 50% of households belong to ABC1 (hereinafter "wealthy block"). We use "wealth" as shorthand for this broader measure of socioeconomic status. In the average block, 27% of households are ABC1 (Table 2). Supplemental Figure E.2 in online Appendix E maps the wealth variable. As a complementary proxy, we treat the six most affluent municipalities (Vitacura, Providencia, Las Condes, Lo Barnechea, La Reina, and Ñuñoa), Santiago's historic "high-income cone," as wealthy.

We rely on baseline (2002) measures of homeownership and wealth because they are predetermined relative to the subsequent regulatory changes analyzed during 2002–2019. Using contemporaneous neighborhood characteristics could introduce reverse-causality concerns, as zoning changes themselves may affect neighborhood composition through redevelopment, residential turnover, housing prices, or new residential construction. Accordingly, homeownership and wealth variables should be interpreted as baseline neighborhood characteristics associated with subsequent regulatory trajectories rather than contemporaneous measures of residents' political preferences or mobilization.

We also include other variables, such as the prevalence of single-family homes. Table 2 presents block-level descriptive statistics, showing an average homeownership rate of 0.71.

Qualitative interviews. To complement our quantitative analysis, we conducted 19 semi-structured interviews from October 2023 to November 2024 with stakeholders such as local and national officials, urban planners, developers, consultants, and non-governmental organizations (NGOs) representatives. We purposively sampled participants to reflect diverse perspectives aligned with emerging quantitative patterns.

The sample was balanced across varying socioeconomic contexts within Santiago, allowing us to examine how private actors participate in land use regulatory processes. For example, we interviewed officials from affluent and low-income municipalities, providing insights into regulatory dynamics across these groups. Supplemental Appendix D provides details about the interviewees' profiles. The interviews enhanced the quantitative analysis and helped refine key variables.

Quantitative research methods: Wealth, homeownership, and changes in height regulation

We analyze the relationship between homeownership, socioeconomic status, and the likelihood of land use regulatory changes using the linear probability model described in Supplemental Appendix A, equation (1). Because spatial concentration of wealth is central to our hypotheses, we use an indicator equal to 1 if the lot is in a wealthy area in 2002, based on the two proxies described above. Another key variable is the share of owner-occupied dwellings in each block. Proximity to the subway also shapes land values and development incentives, because metro access is capitalized into housing prices (López-Morales et al. 2023; Agostini and Palmucci, 2008) and homeowners near stations may have stronger incentives to support upzoning—or weaker incentives to support downzoning—because transit accessibility raises the expected returns from densification. We therefore include distance to the 2002 subway network among the predetermined controls, alongside municipality fixed effects to control for unobserved neighborhood factors.

Qualitative research methods: The role of wealth and homeownership

We conducted in-depth interviews using a structured protocol to understand the factors associated with changes in land use regulation. The interviews focused on three main themes:

- The motivations behind zoning changes.
- The social, economic, and political factors associated with these changes.
- The role of public and private actors in these processes.

We transcribed and anonymized all interviews and coded them systematically in Dedoose, identifying recurring themes in how public and private actors interact within Chile's land use planning framework. Coding underscored the prominent role of wealth concentration and organized residents in regulatory processes.

Results

Quantitative results: Homeowners as a critical and heterogeneous stakeholder

Table 3 presents estimates of equation (1), which examines whether wealth and homeownership are associated with changes in land use regulations. Panel A uses location within the six wealthiest municipalities of Santiago as a wealth proxy. Accordingly, lots in any of these six municipalities are classified as "wealthy" in our regressions and all other lots as non-wealthy. Panel B instead uses block-level wealth from the 2002 Census. Each column represents a separate regression. Columns (1)–(4) take downzoning as the dependent variable, while (5)–(8) upzoning. All models control for the logarithm of the block's population and distance to the 2002 metro network and use robust standard errors clustered at the block level. Panels B-D add municipality fixed effects to control for unobserved local government-level variables. Supplemental Appendix F plots the Panel B estimates.

The quantitative results reveal a positive association between wealth and downzoning, concentrated in affluent blocks with low homeownership rates, that is, where renters are the majority. The sample-average baseline probability of downzoning ranges from 6.6 percentage points in 2007 to 17.9 in 2019 (Table 3), reflecting a citywide trend toward more restrictive regulation.

As wealth is interacted with homeownership, the wealth coefficients in Table 3 are not unconditional associations; each is evaluated at a homeownership rate of zero, almost four standard deviations below the sample mean of 0.71 (Table 2). Comparing them directly with the unconditional baseline probability of downzoning would therefore overstate their implications for a typical block. We instead evaluate the association at representative homeownership rates, the mean and one standard deviation on either side (0.53, 0.71, and 0.89), and report all values in Supplemental Appendix Table F.1.

Evaluated in this way, the positive wealth–downzoning association is substantively meaningful where homeownership is comparatively low. By contrast, at high homeownership rates, the association reverses and can be of comparable size. At the block level (Panel B), the 2007 association is 2.6 percentage points at a homeownership rate of 0.53, about 39% of the 6.6-percentage-point baseline and significant at the 1% level. It is 0.6 percentage points at the sample mean, not statistically distinguishable from zero, and −1.5 at 0.89, about 22% of that baseline and again significant at the 1% level. It changes sign at a homeownership rate of roughly 0.76. The corresponding values for 2019 are 1.6, 0.5, and −0.5 percentage points, none of which is statistically distinguishable from zero, with a sign change at about 0.80.

The coarser municipality-level proxy shows the same pattern, with the reversal occurring at lower homeownership rates. Under this proxy (Panel A), where the wealth indicator does not vary within municipalities and the coefficients are correspondingly larger, the 2007 association is 4.8 percentage points at 0.53, −5.4 at the sample mean, and −15.7 at 0.89, changing sign at about 0.61. All three are significant at the 1% level. Figure F.1 in Supplemental Appendix F plots the block-level association across the full range of homeownership rates together with 95% confidence intervals. As

Table 3. The association between wealth, homeownership, and regulation changes.

	Downzoning				Upzoning			
	(1) 2007	(2) 2011	(3) 2015	(4) 2019	(5) 2007	(6) 2011	(7) 2015	(8) 2019
Panel A: Municipality-level wealth proxy								
(six wealthiest municipalities)								
Wealthy Mun.	35.01***	33.15***	35.10***	28.14***	−21.98*	−17.58	−26.77*	−26.08*
	(3.61)	(3.87)	(4.06)	(5.58)	(11.51)	(12.91)	(13.83)	(13.75)
Homeownership×wealthy Mun	−56.93***	−54.39***	−56.73***	−50.02***	6.39	10.83	15.25	14.56
	(4.90)	(5.24)	(5.50)	(7.53)	(15.84)	(17.78)	(19.07)	(18.94)
Homeownership	17.19***	14.38***	14.16***	13.01**	−1.56	−2.48	−2.40	−3.02
	(1.69)	(2.51)	(2.65)	(5.75)	(12.57)	(14.26)	(15.84)	(15.77)
Observations	19,624	19,624	19,624	19,624	19,624	19,624	19,624	19,624
R-squared	0.06	0.06	0.06	0.06	0.06	0.03	0.05	0.05
Avg Dep. Var	6.598	6.660	7.124	17.85	30.42	38.40	42.16	41.53
Panel B: Block-level wealth proxy								
Wealthy blocks	8.55***	6.89***	5.33**	4.64	5.61	8.30	5.03	5.35
	(2.41)	(2.58)	(2.62)	(2.84)	(4.71)	(5.22)	(4.37)	(4.43)
Homeownership×wealthy blocks	−11.24***	−8.99***	−7.30**	−5.79	1.27	−2.51	−0.70	−0.05
	(3.05)	(3.38)	(3.43)	(3.63)	(6.27)	(7.07)	(6.05)	(6.16)
Homeownership	4.56***	2.74	3.03	5.63***	−3.36	−1.01	−2.23	−4.05
	(1.36)	(2.22)	(2.25)	(1.73)	(2.62)	(2.90)	(2.58)	(2.72)
Observations	19,624	19,624	19,624	19,624	19,624	19,624	19,624	19,624
R-squared	0.57	0.57	0.56	0.76	0.61	0.64	0.73	0.71
Avg Dep. Var	6.598	6.660	7.124	17.85	30.42	38.40	42.16	41.53
Panel C: Block-level wealth proxy—lots in blocks with above-median single-family housing share								
Wealthy blocks	5.31***	3.08	3.35	4.64***	9.98	14.30	6.71	6.24
	(1.52)	(2.16)	(2.16)	(1.63)	(7.38)	(9.17)	(7.40)	(7.44)
Homeownership×wealthy blocks	−7.33***	−4.30	−5.02*	−5.52***	−2.20	−7.56	−1.38	−0.48
	(1.85)	(2.94)	(2.96)	(2.10)	(8.92)	(11.04)	(8.84)	(8.90)
Homeownership	3.80***	1.44	2.57	4.66***	−0.04	4.96	1.64	−0.28
	(1.24)	(2.43)	(2.48)	(1.72)	(2.74)	(3.02)	(2.64)	(2.79)
Observations	15,832	15,832	15,832	15,832	15,832	15,832	15,832	15,832

(continued)

Table 3. (continued)

	Downzoning				Upzoning			
	(1) 2007	(2) 2011	(3) 2015	(4) 2019	(5) 2007	(6) 2011	(7) 2015	(8) 2019
R-squared	0.74	0.73	0.71	0.82	0.66	0.69	0.76	0.75
Avg Dep. Var	5.750	5.850	6.215	17.10	30.87	39.19	43.04	42.30
Panel D: Block-level wealth proxy — lots within large-scale regulatory transformations (> 5 km^2)								
Wealthy blocks	3.32*	2.94*	1.16	4.40*	3.14	6.47	3.20	4.26
	(1.82)	(1.77)	(1.90)	(2.55)	(4.70)	(5.36)	(4.37)	(4.32)
Homeownership×wealthy blocks	−5.59**	−5.28**	−3.32	−5.65*	4.45	−0.23	1.61	0.58
	(2.34)	(2.28)	(2.44)	(3.19)	(6.15)	(7.17)	(6.01)	(6.05)
Homeownership	6.03***	6.06***	6.34***	6.23***	−1.63	−0.19	−1.27	−1.35
	(1.42)	(1.41)	(1.47)	(1.69)	(2.63)	(2.94)	(2.58)	(2.71)
Observations	18,595	19,207	19,194	18,025	18,595	19,207	19,194	18,025
R-squared	0.67	0.68	0.67	0.76	0.69	0.68	0.76	0.74
Avg Dep. Var	5.602	5.426	5.900	14.63	28.75	38.29	42.13	42.54

Note: This table presents OLS estimates of the relationship between wealth and homeownership and the probability of downzoning (columns 1–4) and upzoning (columns 5–8). Dependent variables are multiplied by 100 and should be interpreted in percentage points. Each column reports a separate regression. The unit of observation is a lot in 2002, with regulation changes measured between 2002 and the indicated year. All specifications include log block population and distance to the 2002 metro network as controls. Panels B, C, and D additionally include municipality fixed effects. Panel C restricts the sample to lots in blocks with an above-median share of single-family housing. Panel D restricts the sample to lots within areas undergoing regulatory changes spanning at least 5km^2. Robust standard errors are clustered at the block level (or municipal level in Panel A) in parentheses.

*$p < 0.1$. **$p < 0.05$. ***$p < 0.01$.

discussed below, these estimates are most precise in the earlier years and with finer clustering.

Our data show that the relationship between wealth, homeownership, and land use regulation varies meaningfully across urban contexts. At homeownership rates around one standard deviation below the sample mean, the association between high-income blocks and downzoning is positive and statistically significant in the earlier years of the period. At the sample mean, it is small and not statistically distinguishable from zero in any year. Where homeowners constitute the large majority, it turns negative, remaining statistically significant through 2015, suggesting that local land values and redevelopment opportunities may condition the regulatory outcomes associated with tenure structure. Supplemental Appendix Figure F.1 illustrates this variation. In non-wealthy blocks, by contrast, homeownership is positively associated with downzoning in every year, significantly so at the start and the end of the period.

As baseline measures, wealth and homeownership capture pre-existing neighborhood characteristics rather than residents' contemporaneous political behavior, and may also reflect unobserved differences in redevelopment opportunities, building typologies, market segmentation, turnover, or local organization.

In our main specifications, the associations between neighborhood socioeconomic composition and upzoning are less consistent than those for downzoning. In Panel B, columns (5)–(8), the block-level wealth coefficients are not statistically distinguishable from zero, and the marginally significant negative coefficients in Panel A for the municipality-level wealth proxy are not mirrored at the block level. Given baseline upzoning probabilities of roughly 30–42 percentage points, upzoning appears less systematically associated with neighborhood composition than downzoning in these pooled specifications. As we discuss below, however, this pooled pattern is not uniform: once baseline regulation is accounted for, the block-level specification does show a positive association between wealth and upzoning, and disaggregating by baseline regulatory regime yields associations of opposite sign across regimes (Supplemental Appendices H and J).

A possible concern regarding the association between wealth and downzoning is that wealthy homeowners may have left the area to make way for new buildings, displaced by wealthy tenants entering apartments produced by earlier upzoning. Two reasons suggest limited bias. First, such moves would likely occur within the six wealthy municipalities, yet our results also hold under the municipality-level wealth proxy. Second, the spatial distribution of wealth in Santiago is highly persistent: of roughly 4500 lots in wealthy blocks in 2002, about 3500 were already in wealthy blocks in 1992.

Our findings are robust across several alternative specifications and sample restrictions: retaining only municipality fixed effects as controls; replacing baseline covariates with their 1992 values to mitigate reverse-causality concerns; restricting downzoning to height or FAR decreases of at least 10% relative to baseline; and restricting the sample to numeric-to-numeric regulatory changes, which addresses transitions from unlimited to numeric regulations that may reflect formalization rather than substantive tightening (Supplemental Appendices G and J).

We also examine whether the associations we find are independent of pre-existing regulatory conditions. Indeed, wealthy blocks had a lower baseline FAR in 2002 (0.83 vs 1.25 in non-wealthy areas; Supplemental Appendix Table K.1), indicating they were relatively more restrictive at the outset. Thus, we re-estimate our main specifications

with initial regulation included as a control, using both the block-level and municipality-level wealth measures (Supplemental Appendix Tables H.1 and H.2, respectively). Baseline regulation is clearly consequential, but our main downzoning results are largely robust to its inclusion under both measures. The upzoning results are less stable: the block-level specification yields a positive association between affluence and regulatory relaxation that the municipality-level specification does not reproduce. To examine pre-existing regulation more closely, we disaggregate this variable to assess how the margin available for regulatory change depends on where a neighborhood starts (Supplemental Tables H.3 and H.4). The association between neighborhood demographics and downzoning is stable across measures and specifications. By contrast, the corresponding association with upzoning is more difficult to interpret. The analysis indicates a relationship between affluent blocks and upzoning, but the pattern appears to hold outside the six most affluent urban municipalities of Santiago.[4]

Statistical precision is sensitive to the clustering level. Supplemental Appendix Table G.2 reports inference using standard errors clustered at the 158 regulatory polygons and municipality-level wild-cluster bootstrap p-values. Point estimates are unchanged, but uncertainty increases: under this more conservative inference, only the 2007 interaction estimate remains marginally significant, while later estimates are not statistically distinguishable from zero.

Across all specifications, the spatial concentration of wealth is associated with a higher likelihood of downzoning where homeownership is low. The association weakens as homeownership rises (at the sample mean it is statistically indistinguishable from zero at the block level and already negative under the coarser municipality-level measure) and reverses in affluent areas where homeowners are the large majority. By contrast, in non-affluent zones, homeownership is positively associated with downzoning in every year, significantly so at the start and end of the period.

Supplemental Appendix I examines continuous changes in FAR and does not show robust associations between wealth–homeownership and the magnitude of regulatory changes, suggesting that the main pattern operates primarily through where changes occur rather than through their intensity.

Qualitative results: Zoning as an institutional process open to social and political tensions

This section examines the role of private interests in zoning plans and local ordinances using qualitative evidence collected for this study. We analyze the interactions between private groups and public actors, as well as the strategies through which different actors participate in land use regulatory processes in residential areas.

Developing local ordinances and zoning plans has become increasingly complex, as successive legal reforms added public actors and regulatory requirements. Nearly all interviewees emphasized that implementing substantial changes is slow and fraught with social and political tensions.

While land use regulation involves policy and technical considerations, it remains fundamentally political. The mayor and Municipal Council, elected by district residents, lead the process. Additional public actors, such as environmental agencies, may play roles, but final decisions rest with the Municipal Council, whose members align with the interests of their electoral constituency.

Interviewees highlighted that zoning proceeds smoothly when technical aspects align

with political priorities, whereas misalignment produces contentious debates in which political interests often prevail. One Ministry of Housing and Urbanism official, for instance, described a plan completed in under 3 years thanks to unusually close alignment between the municipality's technical team and the Council (Interview, October 11, 2023).

The technical dimension often refers to the urban planning department and related officials assessing the municipality's territorial needs and identifying the regulatory changes needed to address them and achieve specific social and urban outcomes. However, these policy assessments and recommendations sometimes conflict with council members' perceptions of their constituents' priorities. For instance, while planners may recommend additional development to meet identified needs, council members—responding to residents' concerns—may prioritize regulatory stability. A municipal planner explained:

> If I had only listened to the residents, I would have had to leave the entire municipality as a two-story area. (Interview with a local official from a low-income municipality located in north-west Santiago, October 10, 2023).

Two primary interest groups emerged repeatedly in discussions surrounding land use regulation: current residents and the real estate industry. Residents include both renters and homeowners, with the latter group divided between those interested in maximizing property values through redevelopment and those prioritizing neighborhood preservation.

Most interviewees described residents, and homeowners in particular, as highly prominent participants in land use regulatory debates, but the positions they expressed varied with their interests as property owners. Some noted that homeowners perceive opportunities to obtain substantial gains by selling their land to developers and therefore support more flexible regulations, since higher allowable densities increase land values. Others emphasized that zoning debates are shaped by homeowners seeking to preserve the physical and social character of their neighborhoods, who oppose regulatory changes associated with unwanted development. The following quote illustrates these contrasting perspectives:

> It's really like two extremes. On one side, there's the view of the neighbor who wants to keep living in the area, who doesn't want it to be destroyed. But there's also the view of the neighbor who wants to sell, who wants to profit from their land. So, they say, 'allow more height,' not necessarily out of selfishness, but because everyone has the right to see the community grow. They see vertical development as a form of progress. But then there are others who see real estate development as the breaking apart of how neighborhoods are traditionally formed. (Interview with a local official from a middle-income municipality located in central-west Santiago, October 11, 2023).

Given these two divergent interests among homeowners, the qualitative evidence is broadly compatible with the quantitative patterns above, though not direct evidence of mobilization by tenure type. In affluent areas with high redevelopment values, some interviewees described homeowners as more open to regulatory flexibility that could raise future sale value, while many others prioritized neighborhood preservation and stability. Land-use conflicts in Santiago thus involve heterogeneous and sometimes competing interests among residents rather than clearly identifiable tenure-based blocs.

The interviews also suggest that residents participate in regulatory processes through

formal and informal interactions with council members, who are often attentive to their constituents' concerns.

Interviewees consistently pointed to two factors behind residents' capacity to participate effectively: financial resources and organized advocacy. Higher-income residents were described as having greater access to networks that facilitate interactions with decision-makers, and residents organized through formal associations appeared better positioned to advance their agendas.

Local officials described neighborhood associations as highly experienced participants in advocacy and zoning debates, often engaging directly with council members to express support for particular regulatory positions. Interviewees also suggested that recent legal reforms mandating public participation may have increased the visibility and involvement of organized residents in land use discussions, particularly in debates surrounding dense real estate development. One local official noted:

> The ones who exert the most pressure are the residents and the social leaders. Here, at least, it's the social leaders of the neighborhood associations in the area who want or don't want certain things. They are the ones who exert the most pressure and are very experienced leaders who know how this works. So, they approach council members, lobby them, and apply pressure. (Interview with a local official from middle-income municipality located in central-west Santiago, December 11, 2023)

A local official highlighted the challenge of processing zoning plans in high-income districts such as Las Condes or Lo Barnechea, where residents are typically better informed and more vocal (Interview, December 1, 2023).

Neighborhood-association leaders likewise stress their capacity to organize and to stop proposals that do not reflect their interests. Our qualitative findings thus point to a socially differentiated landscape of land use regulation.

Heterogeneity

We now return to the quantitative analysis to examine three heterogeneity hypotheses drawn from theory and informed by our interviews. The first, drawn from the U.S. literature, posits that resistance to regulatory change tends to be stronger in areas dominated by single-family housing (Ellickson, 2020). In the United States, such low-density regulations have often proven highly persistent over time, generating significant policy and academic debate due to their implications for housing affordability and urban development (Manville et al., 2020). However, our qualitative evidence suggests that residents in Santiago may adopt a more reactive stance, often supporting downzoning after dense and tall buildings are built nearby.

To examine this hypothesis, we re-estimate the main specification using only blocks with an above-median share of single-family housing (Table 3, Panel C). The coefficients are broadly consistent with those in Panel B but generally smaller, suggesting that the association between wealth and downzoning is also present in predominantly single-family areas. At representative homeownership rates, it follows the same profile as in Panel B: 1.4 percentage points in 2007 at a homeownership rate of 0.53, significant at the 5% level; not statistically distinguishable from zero at the sample mean; and changing sign at about 0.72 (Supplemental Appendix Table F.1). As in the baseline specification, higher homeownership is associated with a weaker wealth–downzoning association, while in less affluent neighborhoods, homeownership remains positively associated with downzoning. These findings are consistent with evidence from the United States linking low-density neighborhoods to

stronger support for preserving existing land use regulations.

The second hypothesis, also drawn from the U.S. literature, is that zoning changes affecting large urban areas may create a more balanced representation of affluent residents, homeowners, and the real estate industry, since the industry is more engaged when the economic stakes are larger (Hills and Schleicher, 2015). This is consistent with our interviews: municipal council members were frequently described as responsive to residents, but for large-area changes, interviewees suggested that residents' interests may compete with those of other actors, including the real estate industry.

To test this hypothesis, we analyzed a subsample that includes only large-scale zoning changes, specifically those affecting areas larger than 5 square kilometers. Table 3, Panel D, shows that the pattern changes rather than simply weakening in this subsample. The positive association found at comparatively low homeownership is not clearly present here. At a homeownership rate of 0.53, the estimates are small and not statistically distinguishable from zero in any year. At the sample mean and above, the association is instead negative and statistically significant through 2015, ranging from -0.6 to -1.2 percentage points at the mean, or 11% to 20% of the correspond- ing baseline probabilities (Supplemental Appendix Table F.1). Here too, the wealth–downzoning association declines as homeownership rises, significantly so in every year except 2015. By contrast, the relationship between homeownership and a higher likelihood of downzoning in non-affluent areas remains both statistically significant and consistent in magnitude.

Our third hypothesis is that "floor effects" drive our findings: if baseline regulated density is already low, there is little room left for downzoning, so downzoning should concentrate where it remains feasible. Supplemental Appendix Tables H.3 and H.4 show that areas with higher baseline regulated FAR do drive the association between wealth–homeownership and downzoning.[5]

Discussion and conclusions

This paper examined patterns in the political economy of land use regulatory change in Santiago, Chile, combining a novel dataset georeferencing zoning regulations for residential lots across all urban municipalities over nearly two decades with qualitative interviews of key urban actors. Land-use regulatory patterns are closely associated with socio-territorial inequalities and the differentiated interests of urban residents.

Downzoning and upzoning show markedly different associations with neighborhood socio-demographics. Affluent areas are more likely to experience downzoning, though this is estimated with considerable uncertainty at coarser clustering levels and weakens, sometimes reversing, as homeownership rises. Outside affluent areas, homeownership is positively associated with stricter regulation every year. Homeownership is therefore not associated with a uniform regulatory pattern: its association with downzoning varies with local wealth and with the scope for gains from urban transformation. On average, we find no systematic association between baseline wealth or homeownership and upzoning, but this average masks associations that change sign across initial regulatory conditions and results that differ between our two wealth measures.

Our heterogeneity analyses add context. Wealthy Santiago areas already had lower FAR in 2002, suggesting many affluent neighborhoods were relatively protected from dense development beforehand, and the wealth–downzoning association concentrates where baseline regulation still left room for further restrictions. Where

redevelopment remains attractive, and rules already permit relatively dense building, affluent homeowners may favor additional restrictions less than affluent renters or preservation-minded residents. Future research incorporating redevelopment activity, property transactions, or housing production could clarify the mechanisms.

These findings resonate with a broader Latin American urban literature on the region's distinctive socio-spatial segregation. Unlike the racial and ethnic cleavages that historically structured residential sorting in North American and European cities, segregation in Latin American cities has been predominantly socioeconomic, with upper-income groups concentrated centrally and lower-income populations in the periphery (Roberts and Wilson, 2009; Sabatini, 2003). The expansion of housing markets and subsidy-based housing policies has intensified this income-based sorting, producing a new geography of segregation at a finer spatial scale (Gil, 2019; Monkkonen, 2012; Sabatini, 2003; Sabatini et al., 2001). The patterns we document fit this trajectory: if upper-income groups have long shaped their urban environments through markets and state policy, it is unsurprising that this influence now extends to land use regulation.

These findings also speak to debates on urban inequality, asset accumulation, and land use governance. Much recent U.S. work examines the political economy of regulation through the "homevoter hypothesis," which holds that homeowners support restriction to protect collective goods in which they have a direct stake: limiting supply sustains property values, manages congestion and nuisances, preserves local amenities, avoids redistributive transfers to lower-income entrants, and controls neighborhoods' social composition (Fennell, 2009; Fischel, 2001; Hilber and Robert-Nicoud, 2013; Monkkonen and Manville, 2024; Trounstine, 2023). High-income residents and homeowners are most consistently associated with these preferences and have increasingly displaced the development industry as the key actors shaping outcomes, at least in high-demand cities (Been et al., 2014; Einstein et al., 2019; Gabbe, 2018). As Brouwer and Trounstine (2024, p. 178) note, more research is needed on how community demographics relate to regulatory outcomes. Our results suggest that in highly unequal, segmented cities like Santiago, homeowners are not homogeneous: socioeconomic status conditions their incentives in ways tenure alone does not capture. Wealthier homeowners may gain more from permissiveness than restriction where redevelopment is attractive and baseline rules already allow dense building, since asset value depends not only on excluding externalities but on capturing redevelopment gains. Outside wealthy areas, homeownership remains tied to restriction, suggesting the homevoter hypothesis's asset-protection logic holds where redevelopment gains are less salient and neighborhood stability dominates.

The Chilean case thus contributes to discussions of asset urbanism and property-based political power: land regulation is not purely technical but embedded in tensions over wealth preservation, capital gains, neighborhood identity, and exclusion. Consistent with qualitative studies of urban conflict in Chile (Colin, 2017, 2021; Gil and Bucarey, 2021; Ubilla-Bravo, 2020), our findings suggest affluent groups may have greater organizational and political capacity to participate effectively in regulatory processes, a dimension of unequal political capacity this literature has left relatively underexplored.

This paper also examines land use governance outside the Anglo-American planning systems and homeowner politics that dominate the literature (Fischel, 2001; Gyourko et al., 2021; Hilber and Robert-Nicoud, 2013; Schleicher, 2013). Land-use institutions

differ substantially across countries and legal traditions (Hirt, 2012). Chile combines relatively formalized planning instruments with strong local political pressures and high urban inequality. Unlike highly discretionary systems, where opposition often emerges project-by-project (Einstein et al., 2019; Hills and Schleicher, 2015), zoning conflicts in Santiago arise within broader regulatory planning processes, where organized residents and politically connected actors are closely associated with outcomes. More broadly, the case illustrates how institutional arrangements and social inequality intersect in ways relevant to debates on housing affordability, segregation, and urban development across Latin America and other unequal metropolitan regions.

Our analysis remains primarily descriptive and correlational, and therefore does not identify causal effects. Future research could further investigate the mechanisms linking neighborhood composition, political organization, and regulatory outcomes, particularly by incorporating electoral behavior, campaign finance, development activity, or property transaction data. Additional work could also explore how land use regulation interacts with housing production, displacement, and long-run spatial inequality.

Overall, the evidence suggests that land use regulation in unequal cities cannot be fully understood through technical planning or generalized homeowner preferences alone. The patterns documented here are consistent with regulatory outcomes being associated with socially differentiated forms of political participation and representation that shape who benefits from urban development and who can resist it.

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Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.*

Supplemental material

Supplemental material for this article is available online.

Notes

1. These associations should be read with two qualifications. First, statistical precision is sensitive to the clustering level: under the more conservative municipality-level inference we report below, only the earliest interaction estimate remains marginally significant, while later estimates are not s distinguishable from zero, though they point in the same direction as our main specification. Second, our quantitative measures capture neighborhood characteristics observed before regulatory change rather than political behavior.
2. The municipalities in our sample are Cerrillos, Cerro Navia, Conchalí, El Bosque, Estación Central, Huechuraba, Independencia, La Cisterna, La Florida, La Granja, La Pintana, La Reina, Las Condes, Lo Barnechea, Lo Espejo, Lo Prado, Macul, Maipú, Ñuñoa, Pedro Aguirre Cerda, Peñalolén, Providencia, Puente Alto, Pudahuel, Quilicura, Quinta Normal, Recoleta, Renca, San Bernardo, San Joaquín, San Miguel, San Ramón, Santiago, and Vitacura. Supplemental Appendix Figure E.1 depicts our regulation data; note that coverage is subject to the sample restrictions described at the end of this section.
3. The disaggregation of each possible regulatory change is detailed in Supplemental Appendix B in Table B.1. Figure C.1 in online Appendix C highlights the correlation between changes in FAR and height, indicating a predominantly positive relationship. Figure E.1 in the Supplemental Appendix E illustrates the spatial distribution of upzoning and downzoning during the study period. It shows that these regulatory changes have not been spatially concentrated, as they have occurred across various parts of the Santiago Metropolitan Area. For example, significant upzoning processes are evident in the city's eastern and southern areas. At the same time, downzoning has been prevalent in central Santiago and the eastern and western regions.
4. Our data records the maximum development permitted on each lot, not the floor area actually built on it. We therefore cannot measure how much unused development capacity remains (the gap between what stands on a lot today and what the rules allow). That gap is what makes redevelopment attractive or unattractive to a landowner, so an identical relaxation in permitted FAR may matter a great deal on an underbuilt lot and very little on one already built close to its limit. Our upzoning results should be read with this in mind: they describe formal regulatory change, and the absence of a pooled association cannot be taken as evidence that redevelopment incentives are absent.
5. We thank one of the reviewers for suggesting this analysis.

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